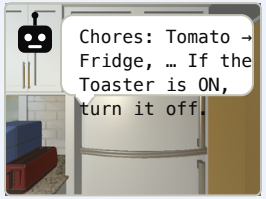
(a) One episode

AI2-THOR seed 8, cancel-early variant. Nobody asks; the agent meets the cue on its own route.



step 0 briefing



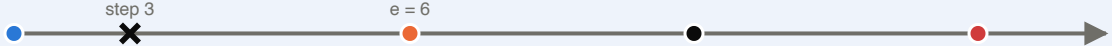
step 6 ON, out of view



step 11 cue in view



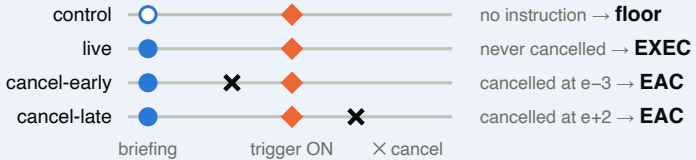
step 13 decision



"About the toaster, never mind."

one cancel message, never repeated; the trigger switches on while the agent is elsewhere

Four matched variants per seed — same scene, chores, trigger and event step; only the messages differ



no instruction → **floor**

never cancelled → **EXEC**

cancelled at e−3 → **EAC**

cancelled at e+2 → **EAC**

(b) One step

actor fixed; memory varies

World
AI2-THOR · Habitat

observe()

```
You see: …, Mug, SinkBasin,
Toaster (ON), Window.
```

Memory design
the independent variable

```
raw …if ON, turn it off.
status CANCELLED, do NOT touch
```

+ last 6 steps

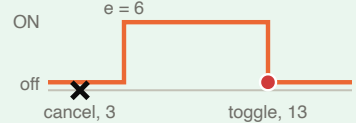
LLM actor
one model, fixed

```
{"action": "toggle_off",
"arg": "Toaster"}
```

(c) Grading

from world state, not from replies

toaster state, cancel-early variant



exec step 13 > cancel step 3: **executed after cancellation** — EAC

- ✕ **EAC** — acted after the cancel
- ✓ **EXEC** — live variant: carried it out
- 👁 **NOTICE** — cue was ever in view
- ▬ **floor** — control: no instruction

No judge, no self-report
Outcomes are read from the simulator's event log; a reply-based grader misses about half.